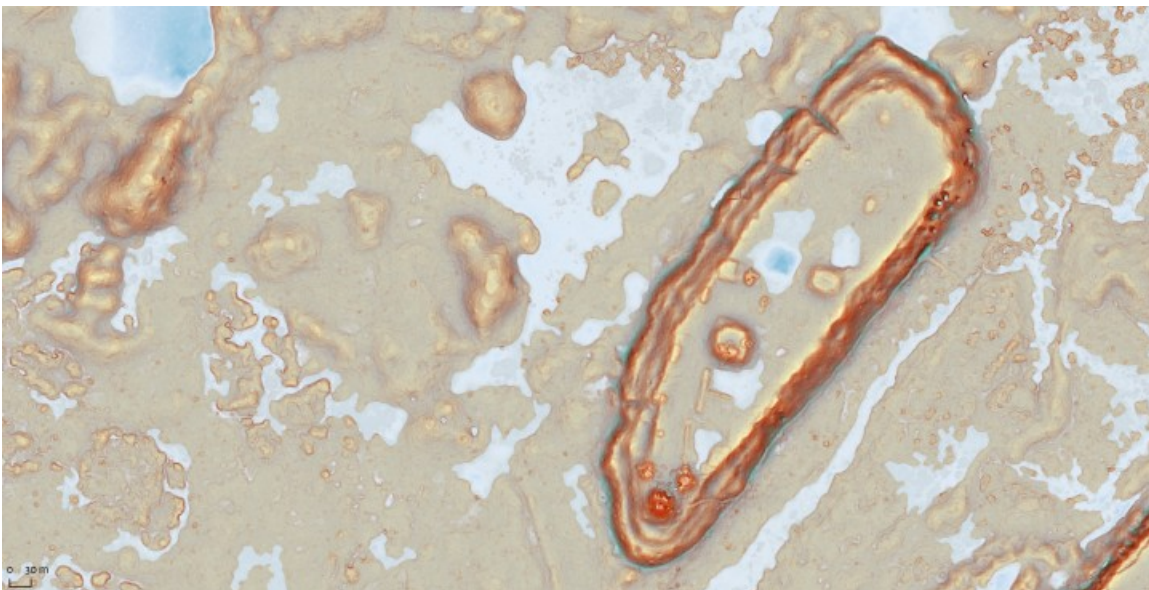


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A Recipe for Simple Red Relief

Luke Auld-Thomas



Walking to the site really is uphill both ways through a swamp.

The Red Relief Image Map (RRIM) has become a popular tool for visualization for archaeological landscapes, especially in the Maya area following its introduction by Inomata et al (2017) at Ceibal, Guatemala. Originally developed by Chiba et al (2008), the RRIM is a composite visualization that represents slope as a chromatic gradient in red, and topographic openness as a brightness gradient. The combined effect is that steep

slopes are vivid red, depressions are dark, ridges are bright, and flat areas are a neutral gray. More recent updates to the RRIM incorporate a second, colorized openness raster, adding a blue-green tint to depressions and bright yellow to ridges while leaving flat areas a neutral gray.

The RRIM has become popular both because it is extremely effective for visualizing and interpreting archaeological landscapes, and because it looks cool—and that’s a not-inconsequential consideration for somebody who’s going to spend 1,000 hours pouring over archaeological terrain visualizations. However, they do have a couple of drawbacks: they require some tinkering for each implementation, both to suppress noise in the computation of the openness rasters and to establish an appropriate upper limit to the slope gradient (i.e., deciding how steep is steep enough to be vivid red). A further complicating factor is that the original RRIM is patented, and while it is unclear exactly what that might mean in legal terms, it’s enough to scare some people off from trying it out.



Simple Red Relief excels at highlighting the micro-topography of karst landscapes

In 2017, I developed my own take on a red relief visualization that draws explicit inspiration from Chiba et al’s original RRIM, but which uses a different “recipe”—avoiding any issues with the patent and producing a visualization that is comparable

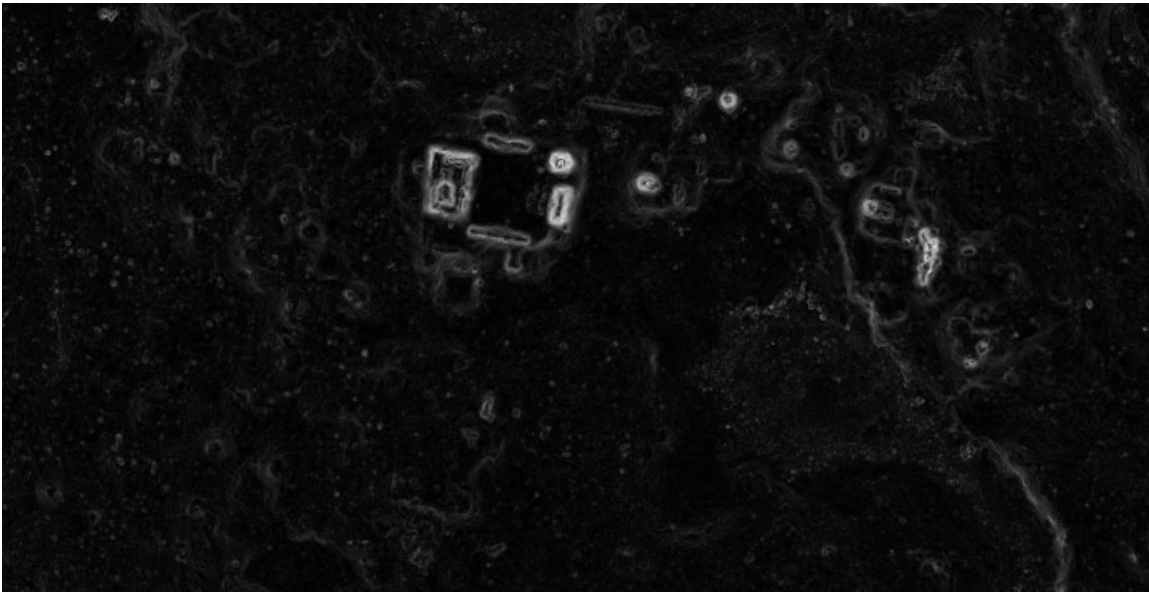
in most respects, with some of its own advantages and disadvantages. I call this visualization Simple Red Relief, because it uses a Simple Local Relief Model (SLRM) in lieu of an openness raster, and this is the visualization that made the rounds in association with Canuto et al's (2018) paper in *Science*, and again last year with Inomata et al's (Inomata et al., 2021) paper on early ceremonial complexes in *Nature Human Behaviour*.

A more detailed post in the future will explore the many effects of tinkering with inputs to this and other composite visualizations: DTM denoising, computation windows, etc. But for now, it seemed high time to go ahead and share the recipe for creating a Simple Red Relief visualization (patent free!), for anyone to use so long as they give this post a shout-out. I hope you enjoy, and if you're pleased (or enraged) with the results, I'd love to see how your own archaeological landscapes look in Simple Red Relief.

Step 1:

Compute the input rasters using the Relief Visualization Toolbox (Kokalj et al., 2011):

- Slope (use default settings, output in degrees)
- Simple Local Relief Model (the radius for trend assessment is in pixels, so will need to be adjusted for the grid resolution of your DTM; I find that setting the radius to the pixel equivalent of 20m works well for most landscapes, 12 gives more detail for flat landscapes, and if you're so inclined you can blend the two for a nice composite effect. But the rest of this example uses 20 m for illustrative purposes)



From such inauspicious beginnings...



For some reason SLRMs always remind me of 1960s spy satellite photos

Step 2:

Layer the slope raster above the SLRM. That's it. That's the step.

Step 3:

Apply color settings to each raster. This is where you will want to do some experimenting to best represent your own data. However, these settings provide a good starting point.

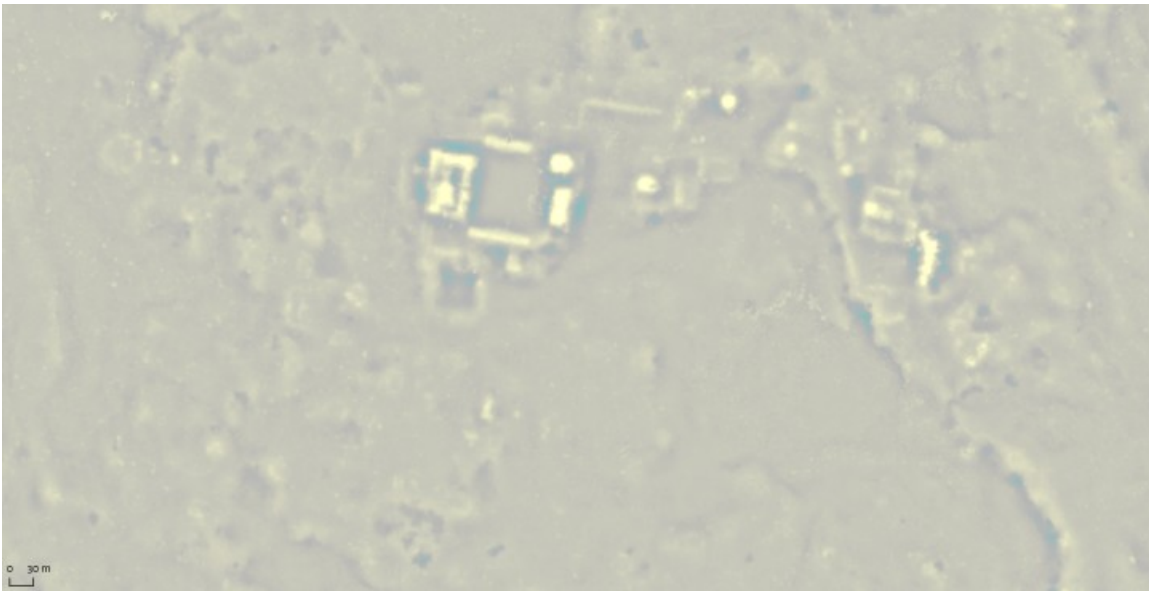
SLRM:

Low Stop: -7.79 rgb(0, 158, 162)

Middle Stop: -1 rgb(138, 138, 138)

High Stop: 8.42 rgb(254, 255, 172)

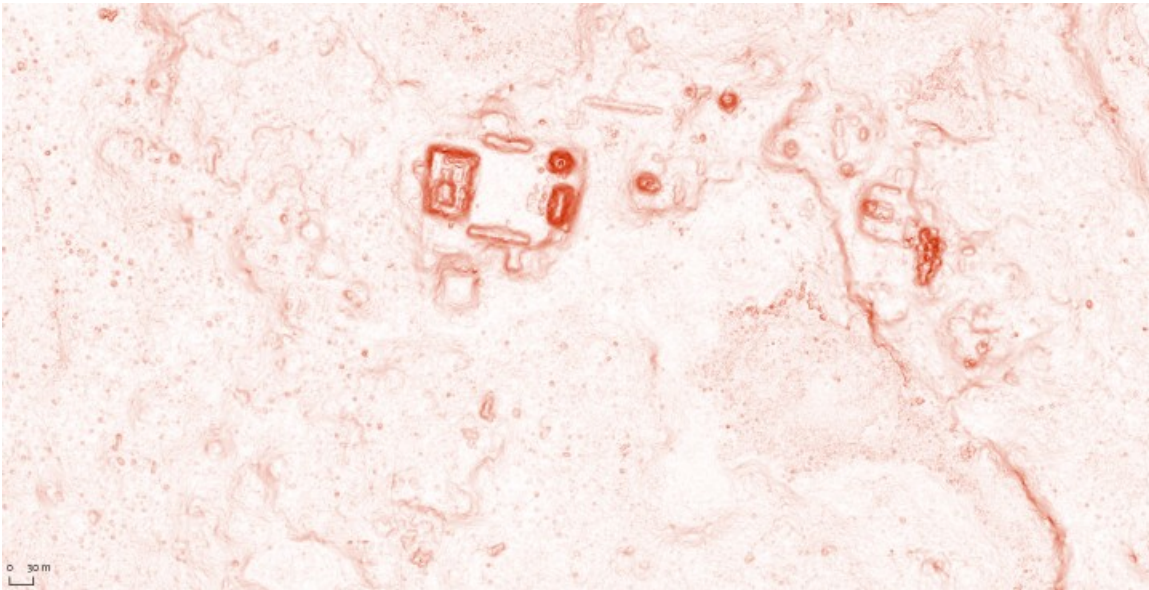
Brightness +40, Contrast +20.



Slope:

Low Stop: 0 rgb(255, 255, 255)

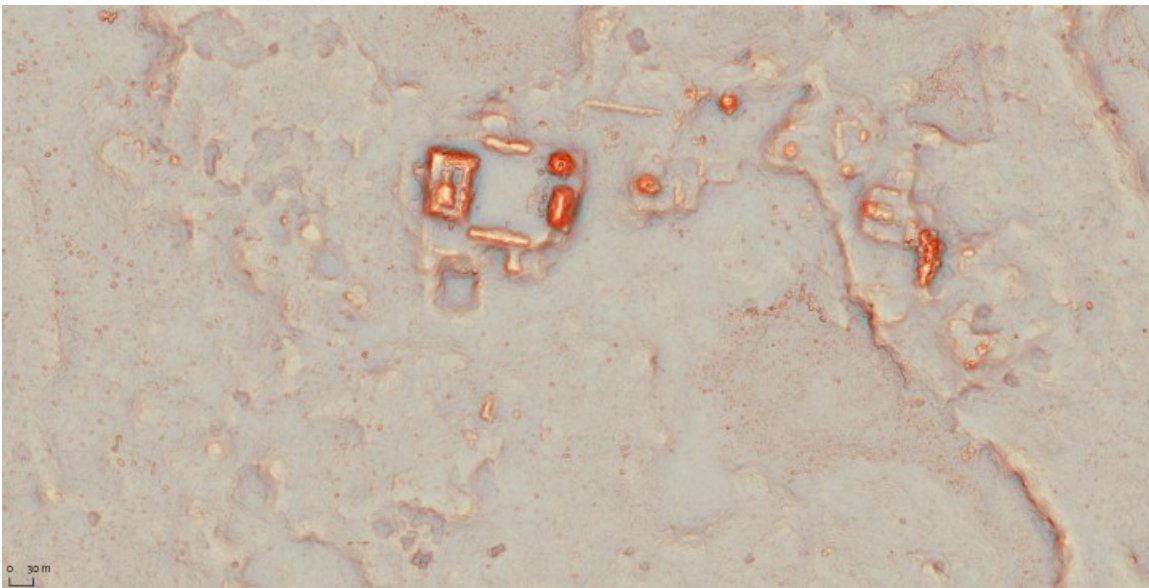
High Stop: 50 rgb(182, 39, 0)



(For those working in QGIS, the color maps are provided as .txt files over on the “Data” page. You can load these in the raster symbology properties dialog to speed the process. Just set the *render type* to “single band pseudocolor” and then click on the little manila folder icon to load the color ramps. You will still need to adjust brightness, contrast, etc. to taste)

Step 4:

Set the blend mode on your slope raster to “multiply” and watch the beauty of your data unfold before you.



Step 5 (Optional):

Since QGIS and ArcGIS Pro both support blend modes, you can stop here and your Simple Red Relief visualization will render on-the-fly as you pan around. However, you might prefer to package the two layers together for greater portability and to reduce on-screen rendering time. If so, you'll need to do it in photo editing software (GIMP, Photoshop, Affinity, etc.). Export both rasters as they appear on your screen (as “rendered images” rather than as raw data), then bring these into your photo editor. Just snap the slope raster on top of your SLRM so that the pixels align, set blend mode to multiply, and you should see an identical visualization to what you previously saw within QGIS/ArcGIS Pro. You can then export this image as a single RGB raster.

Step 6 (Optional):

To bring your new RGB Simple Red Relief image back into your GIS, you need a world file. Create a world file for one of the two input rasters (slope or SLRM), then change the name to match whatever you called the RGB image you just created in your photo editor and put it in the same directory. Now when you bring the RGB image into your GIS project, it will be perfectly georeferenced.

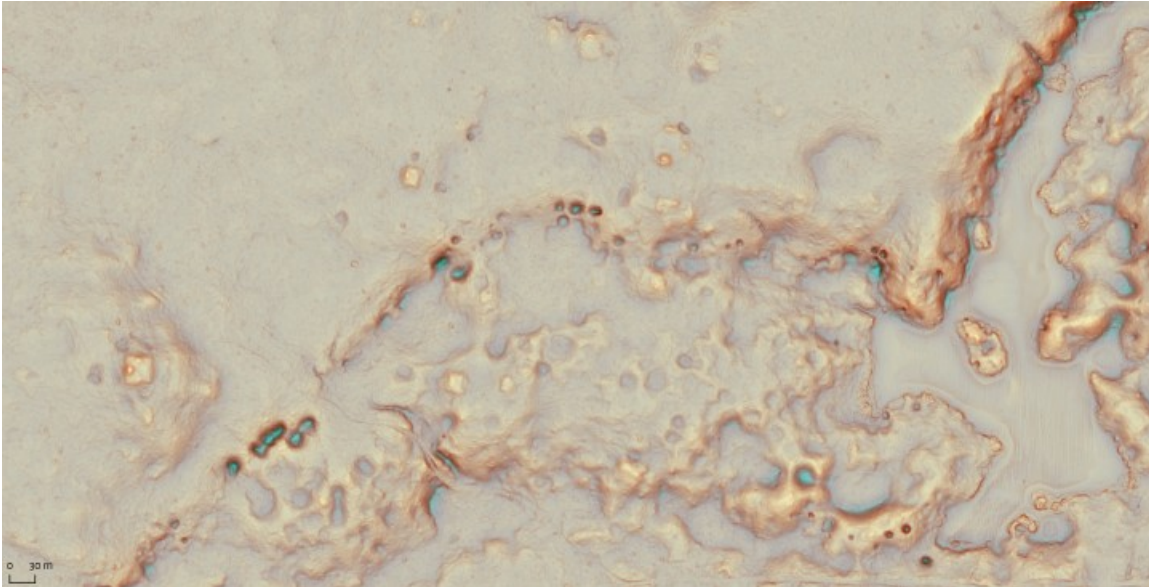
Note for ESRI users: ArcMap and ArcGIS Pro often apply a Gamma adjustment to RGB rasters when they are imported—I'm not sure why or whether this is a setting you can turn off. But it will make your Simple Red Relief look Smurf blue, which is not really ideal. If that happens, just turn the Gamma back down to 1, and it should look how you had it originally.

Showcase:

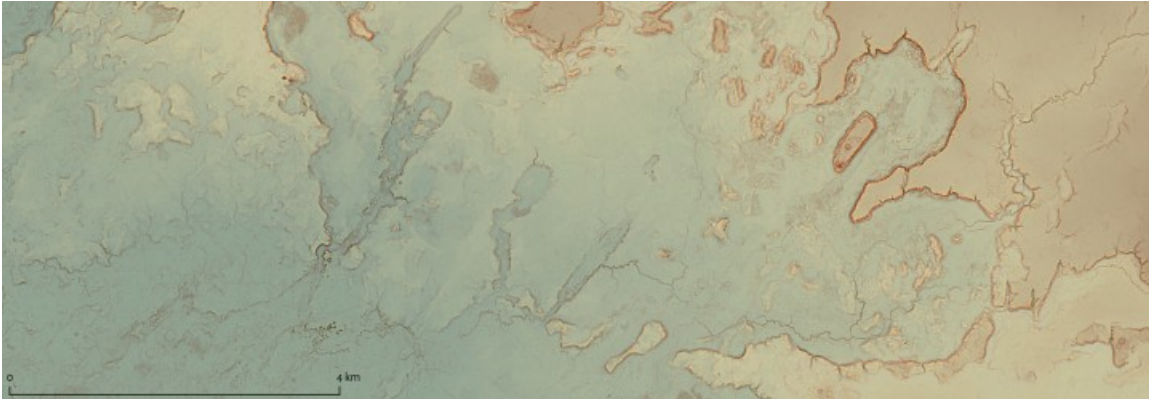
Here are some images from the Corona-Achiotal lidar data and (because why not?) 30m SRTM data for the eastern Peten and western Belize. Note that the nature of the SRTM data—lower resolution, much greater cell-to-cell elevation difference—requires a different set of color ramp parameters for both the SLRM and slope

rasters. This is a good example of the value of puzzling through the color settings to match the data.

Happy mapping!



Small dolines, a smattering of ancient buildings, and a lagoon at far right.



I got yer karst margin plain right here



Raise your hand if you can find the sinkhole at Cival!

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